Observer Dataset Documentation

The **Observer Repository Dataset** is a comprehensive, multimodal clinical encounter dataset designed for healthcare research. It captures real outpatient visits using synchronized video, audio, transcripts, EHR-derived data, and patient and provider surveys. All data are de-identified and structured to support reproducible analysis.

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1. Dataset Overview

The dataset follows an **OMOP-inspired relational structure** and consists of **13 interconnected tables** spanning four primary data modalities:

- Clinical and EHR data
- Multimodal audio and video recordings
- Derived annotations and transcriptions
- Patient and provider survey responses

Each visit is represented by a central `VISIT_OCCURRENCE` record that links patients, providers, clinical data, multimedia files, and surveys.

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2. Data Organization and Directory Structure

The Observer Repository follows a standardized directory layout to organize raw data, derived artifacts, and structured tables.

Top-level Directories

- **Clinic/**: Contains real outpatient clinic visit data.
  - Each visit is stored in its own directory named by the corresponding `visit_occurrence_id`.
  - Visit directories contain de-identified audio, video, annotations, and derived artifacts for that specific encounter.
- **Simcenter/**: Contains Simulation Center videos.
  - These data are collected in simulated clinical environments and are stored separately from real clinic encounters.
- **Tables/**: Contains all structured, OMOP-inspired relational tables.
  - Tables are stored in a standardized format and follow the schema described in this document.

This separation ensures clear provenance between real clinical encounters, simulated data, and structured relational data.

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3. Multimedia and Derived Data

Video and Audio Recordings

Each clinical encounter may include multi-perspective recordings with privacy protections applied.

Characteristics

- Patient, provider, and room perspectives
- De-identified video with face blurring and pose keypoints
- De-identified audio tracks
- Associated transcriptions
- Resolution up to 1080p (MP4 or MOV)

Multimedia files are referenced through the `OBSERVATION` table rather than stored directly in the database.

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3. Core Clinical Tables

PROVIDER

Stores demographic information about healthcare providers.

Columns

- `id` INTEGER: Unique provider identifier
- `year_of_birth` INTEGER
- `gender_source_value` TEXT
- `gender_source_concept_id` INTEGER
- `race_source_value` TEXT
- `race_source_concept_id` INTEGER
- `ethnicity_source_value` TEXT

- `ethnicity_source_concept_id` INTEGER

PERSON

Contains demographic information for patients.

Columns

- `id` INTEGER: Unique patient identifier
- `year_of_birth` INTEGER
- `gender_source_value` TEXT
- `gender_source_concept_id` INTEGER
- `race_source_value` TEXT
- `race_source_concept_id` INTEGER
- `ethnicity_source_value` TEXT
- `ethnicity_source_concept_id` INTEGER

VISIT_OCCURRENCE

Represents an individual outpatient encounter.

Columns

- `id` INTEGER: Unique visit identifier
- `person_id` INTEGER: FK to PERSON
- `provider_id` INTEGER: FK to PROVIDER
- `visit_start_date` TEXT
- `visit_start_time` TEXT
- `visit_source_value` TEXT
- `visit_source_id` INTEGER

NOTE

Stores clinical notes associated with a visit.

Columns

- `id` INTEGER
- `person_id` INTEGER: FK to PERSON
- `provider_id` INTEGER: FK to PROVIDER
- `visit_occurrence_id` INTEGER: FK to VISIT_OCCURRENCE
- `note_date` TEXT
- `note_text` TEXT
- `note_type` TEXT
- `note_status` TEXT

CONDITION_OCCURRENCE

Captures diagnoses recorded during a visit.

Columns

- `id` INTEGER
- `visit_occurrence_id` INTEGER: FK to VISIT_OCCURRENCE
- `is_primary_dx` TEXT
- `condition_source_value` TEXT
- `condition_concept_id` INTEGER
- `concept_code` TEXT

DRUG_EXPOSURE

Records medications ordered or administered.

Columns

- `id` INTEGER
- `visit_occurrence_id` INTEGER
- `drug_ordering_date` TEXT
- `drug_exposure_start_datetime` TEXT
- `drug_exposure_end_datetime` TEXT
- `description` TEXT
- `quantity` TEXT

PROCEDURE_OCCURRENCE

Documents clinical procedures performed or ordered.

Columns

- `id` INTEGER
- `visit_occurrence_id` INTEGER
- `procedure_ordering_date` TEXT
- `name` TEXT
- `description` TEXT
- `future_or_stand` TEXT

MEASUREMENT

Stores quantitative clinical measurements.

Columns

- `id` INTEGER
- `visit_occurrence_id` INTEGER
- `bp_systolic` INTEGER
- `bp_diastolic` INTEGER

- `phys_bp` TEXT
- `weight_lb` REAL
- `height` TEXT
- `pulse` INTEGER
- `phys_spo2` INTEGER

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4. Multimedia Reference Table

OBSERVATION

Stores file references to multimedia and derived artifacts.

Columns

- `id` INTEGER
- `visit_occurrence_id` INTEGER
- `file_type` TEXT - `file_path` TEXT
- `observation_date` TEXT

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5. Survey Tables

PATIENT_SURVEY

Stores patient-reported outcomes and visit experience data.

Responses include general health, technology comfort, satisfaction, Hawthorne effect measures, and open-ended feedback. Most coded responses use ordinal mappings documented per column.

PROVIDER_SURVEY

Captures provider perspectives on visit quality, communication, workload, and technology use, including multiple-choice, Likert-scale, and open-ended responses.

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6. Audit and Vocabulary Tables

AUDIT_LOGS

Tracks access and usage events related to visit data.

Columns

`id` INTEGER

`visit_occurrence_id` INTEGER

`access_time` TEXT

`user_id` TEXT

`workstation_id` TEXT

`access_action` TEXT

Metric-related fields describing event metadata

CONCEPT

Defines standardized clinical concepts derived from controlled vocabularies.

Columns

`concept_id` INTEGER

`concept_name` TEXT

`domain_name` TEXT

`vocabulary_name` TEXT

`concept_class_name` TEXT

`standard_concept` TEXT

`concept_code` TEXT

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7. Notes

- All identifiers are de-identified and non-reversible
- Multimedia files are stored externally and referenced by path

- The schema is designed to align closely with OMOP CDM while supporting multimodal research use cases